Why observable-level twist averaging cannot remove the winding artifact — and where it would work

`twist_qsi_demo` campaign note

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Companion note to `run_twist_average.py`*,* `src/qsi_campaign/twist_average.py`*, and* `outputs/fig_twist_av`
(data: `outputs/twist_average_scan_m{1,2,3}.json`*; spectra cached in* `cache/twist_avg/`*).*
Every number below is an exact-diagonalization result on the cubic 16-site cluster, reproducible from those scripts.

1 The object to be removed

On a torus one unit cell across, the shortest ice-preserving loop that wraps the boundary has *four* sites, while the shortest contractible loop — the hexagon carrying the physical ring exchange — has *six*. Degenerate perturbation theory therefore produces the wrapping process at second order,

$$g_4 = \frac{4J_\pm^2}{J_{zz}}, \tag{1}$$

parametrically *above* the physical third-order scale

$$g_6 = \frac{12\,|J_\pm|^3}{J_{zz}^2}. \tag{2}$$

The low-temperature heat-capacity peak of the bare periodic cluster sits on the g_4 scale: it is an artifact of processes that exist only because the torus is small. Each such process is labelled by its homology winding $\mathbf{w} \in \mathbb{Z}^3$; the census on cubic-16 is given in Table 1. Locally these processes are legitimate ice dynamics — the winding label is the *only* thing wrong with them, so any cure must act on that topological label and on nothing else.

PT order	wrapping loops	winding pattern	$\sum_i w_i$ even	$\sum_i w_i$ odd
2 (4-loops)	36	$24 \times (\pm\mathbf{e}_i),\ 12 \times (\mathbf{e}_i \pm \mathbf{e}_j)$	12	24
3 (6-loops)	80	$48 \times (\mathbf{e}_i \pm \mathbf{e}_j),\ 32 \times (\mathbf{e}_x \pm \mathbf{e}_y \pm \mathbf{e}_z)$	48	32

Table 1: Winding-loop census of the cubic 16-site cluster by perturbative order. The even/odd split of the component sum controls which loops any single twist can annihilate (Sec. 3).

2 What a twist actually does to a winding process

Work in the smooth (translationally invariant) gauge: every $S_i^+ S_j^-$ hop carries $e^{2i\boldsymbol{\theta}\cdot\mathbf{d}_{ij}}$, with $\mathbf{d}_{ij}$ the lifted bond displacement. Because the ice manifold is an emergent-charge-free vacuum, a *completed* ice-to-ice process is sensitive only to the total S^z dipole it transports, and for a loop that dipole is pure topology,

$$\boldsymbol{\rho} = \tfrac{1}{2}\,\mathbf{w}L \tag{3}$$

(verified exactly for all $84 + 120$ winding loops of the cubic and FCC clusters). Three consequences, each verified numerically to machine precision in this campaign:

1. **Contractible loops transport nothing** ($\boldsymbol{\rho} = 0$): the hexagon amplitude is twist-independent. No twist can touch the physics.

2. **Matched-pair interference.** The two virtual orderings of a winding ring exchange transport the dipole in *opposite* half-cell senses ($\pm\boldsymbol{\rho}$), so their smooth-gauge phases interfere *within a single Hamiltonian*:

$$c(\boldsymbol{\theta}) \;=\; g\,\cos(\boldsymbol{\theta}\cdot\mathbf{q}), \qquad \mathbf{q} \equiv 2\boldsymbol{\rho} = \mathbf{w}L \in \mathbb{Z}^3. \tag{4}$$

   This is genuine destructive interference between virtual paths: the amplitude vanishes identically when $\boldsymbol{\theta}\cdot\mathbf{q} = \pi/2 \pmod{\pi}$.

3. **Corner twists are pure gauge.** At $\boldsymbol{\theta} \in \{0,\pi\}^3$ the cosine is ± 1, and a sign on an off-diagonal matrix element is removed by the diagonal unitary $V(\boldsymbol{\theta}) = e^{2i\boldsymbol{\theta}\cdot X}$, $X = \sum_i \mathbf{r}_i S_i^z$. Verified at the full 65,536-state microscopic level: the corner spectrum equals the periodic spectrum to 4×10^{-14}. *The all-π twist does exactly nothing.*

The hoped-for interference therefore exists — but as a cosine of the single projection $\boldsymbol{\theta}\cdot\mathbf{q}$, not as a switch that can be thrown for all $\mathbf{q}$ at once.

3 The two walls

Wall 1: observables average probabilities, not amplitudes. In the solve-per-twist scheme each twist is an independent exact Hamiltonian. Its spectrum is gauge invariant and feels the winding amplitude only through $|c(\boldsymbol{\theta})|^2 \propto \cos^2(\boldsymbol{\theta}\cdot\mathbf{q})$. Averaging $C(T)$ over twists therefore adds *weights*: phases belonging to different twists never sit in the same matrix element and cannot cancel. The uniform torus average leaves every winding loop with

$$\left\langle \cos^2(\boldsymbol{\theta}\cdot\mathbf{q}) \right\rangle_{\boldsymbol{\theta}} = \tfrac{1}{2} \tag{5}$$

— half the contamination power, at every coupling, forever.

Wall 2: direction frustration. Amplitude-level killing by a single twist demands $\boldsymbol{\theta}\cdot\mathbf{q} \equiv \pi/2 \pmod{\pi}$ for *every* winding vector in the census simultaneously. The 24 axis loops require $\theta_i \equiv \pi/2 \pmod{\pi}$ for all three axes; that forces $\theta_i \pm \theta_j \equiv 0 \pmod{\pi}$ — exactly the wrong value for the 12 diagonal loops $\mathbf{w} = \mathbf{e}_i \pm \mathbf{e}_j$. The best any single twist can achieve is

$$\min_{\boldsymbol{\theta}} \sum_{\text{loops}} \cos^2(\boldsymbol{\theta}\cdot\mathbf{q}) \;=\; \frac{12}{36} \;=\; \frac{1}{3}, \tag{6}$$

attained at $\boldsymbol{\theta} = (\pi/2, \pi/2, \pi/2)$: every odd-parity loop annihilated exactly, every even-parity loop at *full* strength. Table 1 shows the same split at third order (48 of the 80 wrapping hexagons are even), so the surviving sector is not a low-order accident: even-winding processes survive *any* phase assignment at *any* order, since $\cos(\text{even} \cdot \pi/2) = \pm 1$.

Proposition 1 (Observable-level floor)**.** *The surviving winding weight of any twist-ensemble average with probability measure* μ*,*

$$W[\mu] \;=\; \sum_{\textit{loops}} \int d\mu(\boldsymbol{\theta})\, \cos^2(\boldsymbol{\theta}\cdot\mathbf{q}), \tag{7}$$

is linear in μ*; its minimum over all probability measures is attained by a point mass at the optimal single twist. Hence on cubic-16*

$$\underbrace{1}_{\textit{bare}} \;>\; \underbrace{\tfrac{1}{2}}_{\textit{uniform average}} \;>\; \underbrace{\tfrac{1}{3}}_{\textit{optimal single twist } (\pi/2)^3} \;>\; \underbrace{0}_{\textit{unreachable}},$$

and no observable-level twist scheme — averaged, optimized, or otherwise — can do better than the single twist $(\pi/2, \pi/2, \pi/2)$*.*

4 What the measurement shows

Table 2 summarizes the 41-point $J_\pm$ scan (full 65,536-state spectra per twist per coupling; see Fig. 1). Every observable-level twist scheme keeps the g_4 *power law* and only shrinks its prefactor $(1 : 0.62 : 0.28)$; only the operator-level mask crosses over to g_6 scaling. The twist protocols do run through the entire dissolved regime ($J_\pm = -0.19$ to -0.30), where the band construction fails; there the single twist is the cleanest tool, because the averaged curve is a mixture whose maximum is grid fragile (at $J_\pm = -0.25$ the $M{=}2$ average puts the peak at $0.051\, J_{zz}$ but $M{=}3$ at $0.086\, J_{zz}$, agreeing with the single twist).

protocol	T_{peak} scaling ($\lvert J_\pm\rvert \le 0.05$, π flux)	T_{peak}/J_{zz} at $J_\pm = -0.05$
bare periodic ED	$\lvert J_\pm\rvert^{1.77}$	0.0130
uniform twist average	$\lvert J_\pm\rvert^{1.76}$	0.0080
single twist $(\pi/2)^3$	$\lvert J_\pm\rvert^{1.80}$	0.0037
winding-free mask	$\lvert J_\pm\rvert^{2.86}$	0.0012

Table 2: Measured lower-peak position: power-law fits and representative values. The twist family reduces the artifact prefactor by at most a factor ~ 3 and never changes the g_4-like exponent; the mask alone reaches the physical g_6 scaling.

5 Why the mask escapes both walls

The winding-free protocol averages the *operator* over the character grid before diagonalizing,

$$H_{\text{clean}} \;=\; \Big\langle V(\boldsymbol{\theta})\, H_{\text{band}}(0)\, V(\boldsymbol{\theta})^\dagger \Big\rangle_{\boldsymbol{\theta}\,\in\,\text{corners}}, \qquad \Big\langle e^{-i\boldsymbol{\theta}\cdot\mathbf{q}} \Big\rangle_{\boldsymbol{\theta}} = \delta_{\mathbf{q},\mathbf{0}}. \tag{8}$$

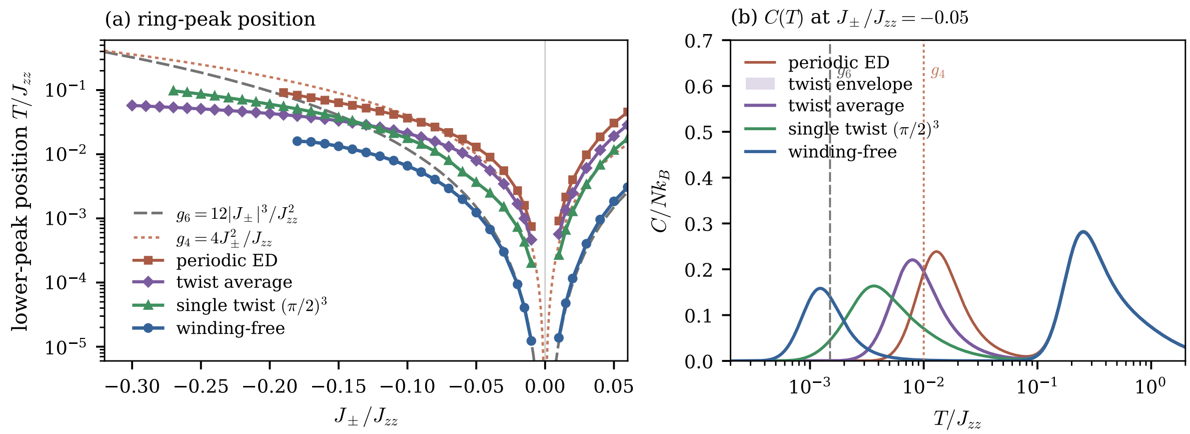

Figure 1: (a) Lower-peak position across the $J_\pm$ scan for the four protocols against the perturbative scales g_4 and g_6. (b) $C(T)$ at $J_\pm/J_{zz} = -0.05$: the twist envelope shows the spread of the individual twisted spectra entering the average; the spinon peak at $T \sim J_{zz}/4$ is identical in all protocols.

That average is coherent — it acts on amplitudes, deleting every winding matrix element identically, at all orders, with no direction compromise (at $M{=}2$ it is the polarization-parity mask of Theorem B). It escapes Wall 1 by acting *before* the nonlinearity of diagonalization, and Wall 2 because character projection uses the full group average rather than one phase point. Homotopy fixes the price: $T^3 = K(\mathbb{Z}^3, 1)$ makes winding the complete topological label, so this cleaning is maximal — and it requires the ice-descended band to exist, i.e. band isolation.

6 Regimes where twist averaging *does* work

The failure is specific to *topological tunnelling amplitudes probed through gauge-invariant observables.* Twist averaging is the right tool when the boundary condition enters differently:

1. **Operator level, always.** But that *is* the winding-free mask: "twist averaging done coherently" and the production protocol are the same object.

2. **Dispersion sampling (Lin–Zong–Ceperley).** When an observable depends on the twist through single-quasiparticle energies $\varepsilon(\mathbf{k} + \boldsymbol{\theta}/L)$ — shell effects, not tunnelling — the twist average literally performs the Brillouin-zone integral and genuinely accelerates convergence to the bulk. Here that regime is the *spinon continuum / high-temperature sector* ($T \sim J_{zz}$), which lives outside the ice manifold, carries no winding contamination, and is where the simulation plan already uses the twist grid legitimately (FTLM $S(\mathbf{q}, \omega)$ interpolation).

3. **Clusters where winding is subleading.** On a torus at least two unit cells across in every direction, the shortest winding loop is longer than the hexagon, the artifact appears beyond order six, and the unremovable $\frac{1}{3}$–$\frac{1}{2}$ residual multiplies something already parametrically negligible. (Not available here: cubic-16 *and* FCC-32 both host winding 4-loops along their short cycles.)

4. **Censuses without the even-parity sector.** If a cluster's winding vectors all had odd component sum, $(\pi/2)^3$ would annihilate the entire leading artifact in amplitude. Cubic-16 fails this (12 even 4-loops), and even-parity windings reappear at higher order regardless — a geometric accident, not a strategy.

Remark. *The dichotomy worth remembering:* twist averaging fixes **k**-quantization errors; it cannot fix topological tunnelling amplitudes*, because gauge-invariant observables see those only through their magnitude. For gauge-sector thermodynamics on ED-sized QSI tori, observable-level twisting — averaged or optimally chosen — reduces the winding artifact by at most a factor of three and never changes its power law; complete removal requires phases to meet* inside *matrix elements, which only the operator-level character projection provides.*